Fiscal Year:	FY 2023	**Task Last Updated:**	FY 02/17/2023
PI Name:	Rice, Kelly Ph.D.		
Project Title:	Assessing the Impact of Agr Quorum Sensing on Staphylococcus aureus Physiology in the Space Flight Environment		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32611-0700	**Congressional District:**	3
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	02/08/2021	**End Date:**	02/07/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:	Freeland, Denise	**Contact Phone:**	321-867-5878
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Dr. Robert Shields (Co-I) moved from the University of Florida to Arkansas State University in 2021, where he is now an Assistant Professor. He will continue to be a Co-I on this grant; UF and ASU will execute a subaward starting in year 2.		
COI Name (Institution):	Carroll, Ronan Ph.D. (Ohio University) Edelmann, Mariola Ph.D. (University of Florida, Gainesville) Nicholson, Wayne Ph.D. (University of Florida, Gainesville) Shields, Robert Ph.D. (Arkansas State University)		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	Effective protection of astronaut health is predicated on a thorough understanding of the effects of the space flight environment on virulence potential of bacterial pathogens. Staphylococcus aureus is a pathogen that can cause a wide variety of devastating infections, many of which result from the ability of S. aureus to grow as biofilm, its wide arsenal of secreted virulence factors, and its highly adaptive metabolism. These properties are regulated, in part, by its Accessory Gene Regulator (Agr) quorum sensing system. Analysis of S. aureus global gene expression data from a previous space flight experiment (Biological Research In Canisters (BRIC)-23; GeneLab Data System (GLDS)-145)) on the International Space Station (ISS) suggested that the Agr-mediated quorum sensing system (QSS) and expression of some of its target regulon (secreted toxins, exoproteases) were highly upregulated during space flight growth. The overall hypothesis of this proposal is that the S. aureus Agr quorum-sensing system is an important regulator of biofilm formation, physiology, and virulence factor expression when this bacterium is exposed to microgravity or other variables related to the space flight environment. To address this, S. aureus biofilm formation in a wildtype and isogenic agr-deficient mutant strain will be compared using a random positioning machine (RPM) to simulate microgravity, and an ISS space flight experiment will be conducted using 4 Biological Research in Canister Petri Dish Fixation Units (BRIC-PDFUs). One BRIC-PDFU (n=5 PDFUs + 1 HOBO temperature logger) will be used to grow a S. aureus transposon library, in order to define which gene(s) are required for thriving in the space flight environment. Two BRIC-PDFUs will be used to grow S. aureus wildtype (same strain used in BRIC-23) and its isogenic agr mutant (n=5 PDFUs + 1 HOBO temperature logger per strain), to determine the contribution of Agr quorum sensing to gene expression changes previously observed in the BRIC-23 flight samples. In a fourth BRIC, PDFUs modified to cancel the potential confounding effects of microgravity on fluid geometry, and hence aeration, in BRIC units (modifications are being developed by Co-Investigator Nicholson through a separate NASA project), will be used to grow the wild-type S. aureus strain (n=5 modified PDFUs + 1 HOBO temperature). The growth media, time, and other experimental variables will be identical to those previously used in the BRIC-23 experiment in order to directly compare results. Asynchronous ground controls for each of the four space flight BRIC-PDFUs (n=5 PDFUs + 1 HOBO temperature logger per BRIC) described above will be treated identically in the ISS Environmental Simulator at Kennedy Space Center (KSC) using the recorded flight temperature data. Upon return to Earth, flight samples and ground-based controls for the wild-type and agr mutant BRIC-PDFUs will be processed for RNA-seq, secreted proteomics, metabolite, and growth measurements. The transposon library BRIC-PDFUs will be processed for DNA isolation, Illumina sequencing, and Tn-seq analysis. These proposed experiments will allow us to make more refined predictions about the pathogenic potential of S. aureus, especially in relation to astronaut health. These experiments align with Decadal Survey Recommendation P1, sub-topics MB-B (Studies designed to determine how spaceflight conditions affect microbial biofilm formation... and their potential to cause disease) and MB-D (to use the ISS as a microbial observatory to determine foundational results... that contribute to the health of future space explorers), as well as address several guiding questions of the 2016-2025 SB Science Plan: To help identify the "underlying genetic, molecular and biochemical processes influenced by the space environment" for S. aureus (MB-1), and offer additional insight into how space flight conditions such as microgravity "...influence microbial biofilm formation and microbial community development" (MB-3).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Agr is a global regulator of both virulence and metabolic genes, therefore stimulation of this quorum-sensing system in response to space flight conditions may impact the pathogenic threat of S. aureus to astronauts during long-term space flight missions, especially since space flight is known to impose a suppressive effect on both innate and adaptive immune cell responses to pathogen-related stimuli. The research proposed in this application is directed at gaining a better understanding of the contribution of Agr quorum sensing to physiological adaptation and pathogenic potential of S. aureus in response to the space flight environment. It is anticipated that the results of these proposed experiments will allow us to make more refined predictions as to which virulence properties and physiological changes are likely to occur in S. aureus during space flight, and how this may impact its pathogenic potential and effect on astronaut health.
Task Progress:	Aim 1: ISS flight experiment to elucidate the physiological response of S. aureus to the space flight environment • In year 2, we completed four pre-SVT tests in preparation for the Science Verification Test (SVT). Protocols were optimized in the first three tests, especially with respect to achieving the target initial inoculum on the plates for the S. aureus TnSeq library, S. aureus UAMS-1, and the isogenic agr mutant strains. Success criteria for the SVT were also fine-tuned in the first 3 pre-SVT experiments. Data from pre-SVT #4 met all the success criteria. • Modified petri dishes have been obtained from the supplier used in the Biological Research in Canister Petri Dish Fixation Units (BRIC-26) study – enough for the SVT, Experiment Verification Test (EVT), flight, and ground control experiments. • SVT was completed February 6-13, 2023. De-integrated samples (frozen at -80 C) have been received by the Principal Investigator (PI) and experiments are in progress to process the samples and assess for success criteria. Aim 2: Determine the effect of Agr and simulated microgravity on S. aureus biofilm growth • We used a previously created Staphylococcus aureus UAMS-1 (UAMS-1) agr P3-GFPuv reporter strain to grow and image biofilms on the random positioning machine (RPM) using 0.9 g and 0 g settings. These results suggested that the agr P3 promoter is much more active when biofilms are grown under 0 g simulation compared to 0.9 g simulation. This is a striking result because these biofilms were grown in a medium that physiologically represses induction of the Agr quorum sensing system. Therefore, a signal in the 0 g simulation appears to be overriding this repression. Efforts in year 3 will focus on verifying and dissecting the mechanism of this regulation. • We have constructed new constitutive superfolder green fluorescent protein (GFP) reporter plasmids in the pMK4 shuttle vector (more stable in strain UAMS-1), as well as promoterless GFP and agr P3-GFP reporter plasmids for use in strain UAMS-1 and its agr deletion mutant. These are all currently being validated for use in RPM biofilm studies described in Aim 2.
Bibliography Type:	Description: (Last Updated: 02/19/2026)
Abstracts for Journals and Proceedings	Davis KAg, Hauserman MH, Rice KC. "Direct imaging of oral streptococcal & Staphylococcus aureus biofilms grown under simulated microgravity." Faculty Poster Session, 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. Abstracts. 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. , Nov-2022

Abstracts for Journals and Proceedings	Hauserman MH, Edelmann M, Nicholson W, Rice KC. "Orientation of the oxygenation membrane on horizontally rotating High Aspect Ratio Vessels (HARVs) impacts Staphylococcus aureus physiology and quorum sensing response." Microbes 1 Oral Presentation. 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. Abstracts. 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. , Nov-2022
Abstracts for Journals and Proceedings	Ocampo J, Rice KC, Edelmann M. "Examination of Enterobacter cloacae physiology and proteome changes during growth in low-shear modeled microgravity (LSMMG)." Graduate Student Poster Session, 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. Abstracts. 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. , Nov-2022